

MID TERM EXAMINATION

Course: Data Structures (CS101)

Set: C | Time: 3 Hours | Max Marks: 128 | Date: 14-03-2026

Total Questions: 17 | Answer all questions.

Name: _____

Roll No: _____

Student Sign: _____

Teacher Sign: _____

SECTION A: MCQs

1. Which of the following is NOT explicitly mentioned as a type of scenario used in architectural evaluation? [3 Marks]
 - a) Use-case scenarios
 - b) Performance scenarios
 - c) Growth scenarios
 - d) Exploratory scenarios

2. A direct consequence of using an off-the-shelf component is that the architecture must primarily: [3 Marks]
 - a) Change the component's internal behavior.
 - b) Adjust to the component's way of working.
 - c) Eliminate vendor dependency.
 - d) Increase customization options.

3. In design decision-making, which factor is guided by what the system must do, such as supporting payments or product search in an e-commerce system? [3 Marks]
 - a) Non-Functional Requirements
 - b) Stakeholder Needs
 - c) Functional Requirements
 - d) Technology Constraints

SECTION A: SHORT ANSWERS

4. Name and briefly explain two types of scenarios mentioned for architectural evaluation. [6 Marks]

5. Explain how using off-the-shelf components can lead to "extra integration work" and "limited customization." [6 Marks]

6. According to the text, what role do "Technology Constraints" and "Team Skills and Experience" play in design decision-making? [6 Marks]

7. List four key stakeholders whose participation is required for successful architectural evaluation. [6 Marks]

8. Describe two negative impacts of off-the-shelf components related to vendor interaction and future changes. [6 Marks]

9. Briefly explain the significance of "Risk and Uncertainty" and "Reusability and Extensibility" as factors for design

decision-making.

[6 Marks]

10. What are "Risks," "Sensitivity points," and "Trade-offs" in the context of successful architectural evaluation? [6 Marks]

SECTION A: LONG ANSWERS

11. Elaborate on the importance of "Clearly Defined Quality Attributes" and "Well-Specified Scenarios" for a successful architectural evaluation, providing examples as given in the text. [11 Marks]

12. Discuss the challenges and limitations introduced by using off-the-shelf components, focusing on how the architecture adapts and potential quality attribute impacts. [11 Marks]

13. Explain how "Functional Requirements," "Non-Functional Requirements," and "Stakeholder Needs" collectively guide design decision-making in software architecture. [11 Marks]

14. Describe the role of "Structured Evaluation Methods" and an "Experienced Evaluation Team" in achieving successful architectural evaluation, including examples of methods and team expertise. [11 Marks]

15. Analyze how "Budget and Schedule," "Team Skills and Experience," and "Technology Constraints" interact as practical factors influencing design decisions. [11 Marks]

16. Drawing from the text, explain how the "Impacts of Off The Self Components on Architecture" might influence "Factors for Design Decision-Making in Software Architecture," specifically considering budget/schedule and customization aspects. [11 Marks]

17. Discuss the comprehensive nature of successful architectural evaluation by explaining how "Architectural Documentation," "Proper Tools and Techniques," and "Good Communication" contribute to the process. [11 Marks]

--- End of Question Paper ---